

VISHAL SONI

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Education

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering | CGPA: 9.01/10

Sep 2023 – Present

Bhopal, India

Experience

Indian Institute of Technology Ropar — VLED Lab

May 2026 – Present

Software Engineering Intern

Remote

- Contributing to open-source educational infrastructure tools under IIT Ropar faculty mentorship; involved across design, implementation, review, and deployment.

Open Source

- eSim (FOSSEE, IIT Bombay)** — Ported eSim EDA to Ubuntu 25.04 by resolving 13 dependency and toolchain failures across KiCad, GHDL, LLVM, Python 3.13, and package repositories.

Projects

CollabCode — Real-Time Collaborative IDE (Next.js, TypeScript, Go, WebSocket, PostgreSQL) Feb – May 2026

- Engineered a low-latency collaborative coding platform with a custom Operational Transform engine to resolve concurrent edit conflicts across distributed sessions.
- Designed a Pub/Sub scaling architecture in Go Fiber handling multi-room WebSocket sessions with JWT-authenticated isolation; integrated Monaco Editor with live code execution.
- Built multi-file workspace management, room-scoped chat, and a structured REST API layer backed by PostgreSQL with session-aware state persistence.

ChronoVault — Distributed Encrypted File Vault (Go, React, FastAPI, AES-256-GCM, IPFS) Nov 2025 – Feb 2026

- Designed the core encryption and storage layer: AES-256-GCM encryption, deterministic file sharding, and Merkle-tree integrity verification for tamper-evident chunk storage.
- Implemented decentralized-style chunk distribution over IPFS with secure file reconstruction, biometric authentication, and cryptographic key derivation workflows.
- Collaborated within a 5-member team, contributing to system architecture decisions, cross-component integration, and technical design across the full stack.

Arvis X — AI-Powered Travel & Booking Platform (Next.js, Cashfree, AWS Lambda, Kubernetes) Sep 2024 – May 2025

- Built a full-stack travel and booking platform supporting buses, trains, flights, hotels, and events with role-based dashboards for users, vendors, and administrators.
- Engineered real-time seat reservation using Redis locking, WebSockets, and distributed cache invalidation; integrated Cashfree payments, wallet, coupon engine, QR e-tickets, and automated refunds.
- Developed AI-powered travel assistant, itinerary generation, and personalized recommendations using LLM APIs; deployed on AWS Lambda and Kubernetes with Terraform-managed IaC.

DineOps — Restaurant Management Ecosystem (Next.js, Cashfree TypeScript, Prisma, Supabase) Jun 2024 – Sep 2024

- Developed a production-grade restaurant management platform with customer, cashier, kitchen, delivery, admin, and super-admin portals using role-based access control.
- Built end-to-end order workflows: POS operations, kitchen display systems, delivery tracking, reservations, inventory management, and AI-powered concierge for personalized dining.
- Designed dual-mode persistence with PostgreSQL production storage and offline JSON fallback; deployed on Vercel and Supabase with JWT auth, real-time notifications, and CI/CD pipelines.

Achievements

- AIR 217** in TCS CodeVita Season 13 Round 1 (100,000+ participants); ranked **3rd at VIT Bhopal** university-wide; final global rank **4,771**.
- Finalist, Smart Delhi Hackathon** — selected from 15,000+ participants for city-scale solution development track.
- Selected as mentee (**top 0.1%** of 80,000+ applicants) for **Codeforces Graphs Camp**; mentored by a Codeforces Master on hard graph problems (shortest paths, traversal, connectivity).
- Finalist, NeoCodeathon – June 2026** — qualified for the Finals Round based on outstanding performance in the Prelims.

Certifications

AWS Certified AI Practitioner

Amazon Web Services

AWS Certified Cloud Practitioner

Amazon Web Services

AWS Certified Solutions Architect

Amazon Web Services

Technical Skills

Languages: Go, Python, C++, Java, TypeScript/JavaScript, PHP, SQL

Backend & Systems: REST APIs, WebSockets, Operational Transform, Serverless, PostgreSQL, MySQL, DynamoDB, Redis

Cloud & Infrastructure: AWS (Lambda, EC2, EKS, RDS, S3, ALB, IAM), GCP (Cloud Run, Firebase), Kubernetes, Terraform

DevOps & Observability: Docker, GitHub Actions, Jenkins, Prometheus, Grafana